

Enhanced grounded capacitor multiplier and its floating implementation for analog filters

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Abstract—A current-mode capacitor multiplier using a differential amplifier with exponential current scaling is presented. The proposed circuit consists of a reduced implementation of a multiplier based on differential amplifiers with decreased need for parameter matching compared to a conventional structure, thus reducing power consumption and silicon area. It includes an exponential current scaling scheme to provide large multiplication factor. Additionally, the differential input stage offers the capability to implement a floating realization for bidirectional applications. The circuit operates with a minimum supply of a gate-source voltage and two saturation voltages of a transistor, making it suitable for low-voltage applications. Experimental results from a fabricated prototype in 0.5 μ m technology show the operation of the grounded and floating realizations, and their performance when implemented in a second-order low-pass analog filter.

Index Terms—Analog Integrated Circuits, capacitor multiplier, low-voltage analog circuits

I. INTRODUCTION

SYSTEM integrability has become very important in the last few decades as the tendency of portable devices to influence daily life is constantly growing. Digital system design has achieved integration of millions of gates in a single die; unfortunately, analog design has faced a significant challenge in this matter due to certain applications, such as analog filters that normally include passive components, i.e. capacitors, which occupy large silicon area. Additionally to portability, integrated systems are increasingly demanded with better performance; this includes high accuracy, reduced distortion, and wide operational frequency range. For this reason ingenious capacitance multiplication schemes have been reported lately to enhance these traits while decreasing the required silicon area.

These circuits can be classified into two types, voltage- and

current-mode multipliers. The first type is based on the well-known Miller effect implemented with a voltage amplifier. The equivalent capacitance is defined as the base capacitance multiplied by the gain factor of the amplifier. Typically, a transistor in common-source configuration is employed as it exhibits large gain and offers signal inversion to obtain a negative feedback path, although more elaborated structures may be utilized [1], [2]. Considering a transconductance gain g_m , and a drain resistance r_o , the multiplication factor provided by the transistor is of the form $g_m r_o$ for which, unfortunately, changes in device parameters, biasing current, and voltage across terminals cause considerable variations, thus significantly compromising the accuracy. Additionally, the output dynamic range is notably reduced due to the high gain. This type of multipliers is normally used in applications where the multiplication factor is permitted to have large variations with a minimum required value. They are commonly used for frequency compensation in two- (or more) stage amplifiers, usually known as Miller compensation.

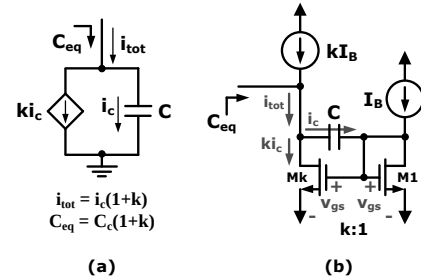


Fig. 1. Current-mode capacitor multiplier showing a) concept of operation and b) practical implementation.

The second type of multipliers, current-mode multipliers, is based on magnifying the capacitive effect by means of current scaling. Fig. 1a shows the concept of operation, where a current in a capacitor C , i_c , is sensed; a current-controlled current source multiplied by a factor k sinks the magnitude $k i_c$ which results in an output current $i_o = i_c(1+k)$, thus exhibiting an equivalent capacitance of $C_{eq} = C(1+k)$. Conventional topologies straightforward based on this scheme offer good accuracy but exhibit limited multiplication factor, decreased effective bandwidth, increased power and silicon area, and only allow single-ended implementation [3]-[5].

In this brief, we propose a reduced topology based on a differential amplifier, which offers the implementation of a floating realization for analog applications. The rest of the paper is divided as follows: Section II-A describes a

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conventional current-mode multiplier and variants describing enhancement in accuracy and multiplication factor. Section II-B describes the proposed circuit including its floating implementation, and frequency analysis is in Section II-C. Section III presents simulation and experimental results of the proposed circuit implemented in a second-order low-pass filter application, and Section IV concludes this work.

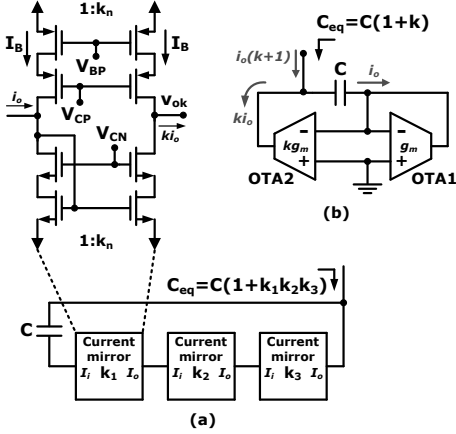


Fig. 2. Current-mode capacitor multipliers based on a) n stages in series for large multiplication factor [6], and b) using differential amplifiers [7].

II. PROPOSED CIRCUIT

A. Capacitor Multipliers

The practical implementation of the basic current scaling scheme is shown in Fig. 1b, where a diode connected transistor M1 senses an ac current in the capacitor, i_c , inducing voltage v_{gs1} which is mirrored to transistor Mk, a scaled replica of M1. For a scaling factor k , Mk sinks a current ki_c , thus producing an output current $i_o = i_c(1+k)$ and resulting in an equivalent capacitance $C_{eq} = C(k+1)$, similar to the scheme in Fig. 1a. Device parameters can be highly matched using proper layout techniques; hence, accuracy is considerably increased. Although it presents a variety of disadvantages; note that the silicon area and power consumption increase proportionally to the gain factor k . Also, for large values of k larger silicon area is required, and parameter matching is compromised in devices located apart from the transistor being replicated, therefore the multiplication factor is considerably limited. The primary non-ideal effects included in this implementation comprise the denominated Equivalent Series Resistance (ESR), and the Parallel Load Resistance (PLR). In Fig. 1b, the ESR is defined by the resistance of the diode connected transistor M1, $1/g_{m1}$, where g_{m1} is its transconductance gain, and it constitutes the high frequency corner of the operational range. The PLR is defined as the drain resistance of M2, r_{o2} , and it constitutes the low frequency corner of the operational range.

Many variations based on this topology have been reported in the past. Such is the case of the multiplier reported by Choi et. al. in [6], shown in Fig. 2a. It consists in cascading n stages in series, which results in a total multiplication factor of $1+k_1 \cdot k_2 \cdot \dots \cdot k_n$. This topology is attractive as the final

multiplication factor can be scaled exponentially with little increase in number of devices and consumed current, offering reduced silicon area and power consumption. Note that all stages are cascoded in order to minimize the drain-source voltage variations and, hence, increase accuracy in current mirroring. In this case the ESR is defined by the resistance seen at the input node of the first stage, given by $\approx 1/g_{mN}$, where g_{mN} is the transconductance gain of the diode-connected input transistor; the PLR is defined as the cascoded output resistance of the last stage.

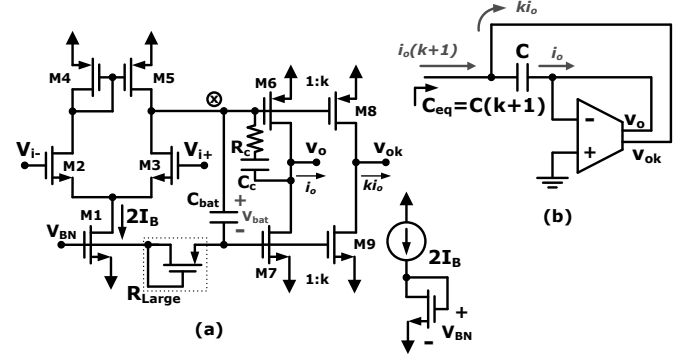


Fig. 3. a) Current scaling scheme using a differential input stage [8], and b) configuration to operate the circuit as a grounded capacitor.

Another topology that offers performance enhancement to the basic current-mode multiplier is that shown in Fig. 2b, reported by Aguado-Ruiz, et. al. in [7], where each of the transistors in the basic topology are implemented using a differential amplifier. OTA1 is connected as voltage follower, presenting the same shunt feedback loop as transistor M1 in Fig. 1b, and similarly, a resistance $1/g_m$. OTA2 offers a scaled transconductance with respect to OTA1, defined as $g_{m2} = k g_{m1}$. Both amplifiers have the same input signals, hence OTA2 provides a current k times larger than that sensed by OTA1. This results in an output current $i_o = i_c(1+k)$ equivalent to the conventional scheme in Fig. 1b. Unfortunately, besides the increased number of devices and power consumption, the offset voltage between the differential inputs in each of the amplifiers due to mismatch in device parameters is inevitable even when using proper layout techniques. In the case of OTA1, the shunt feedback adjusts the output level according to this offset in order to regulate the current in its output devices; however, the induced offset voltage is copied to OTA2, which creates the possibility of mismatched currents between its p- and n-type output transistors, hence increasing an undesired dc level. Equivalent to the conventional structure, the ESR is defined as $1/g_m$ from OTA1, and the PLR as the output resistance of OTA2.

The two amplifiers have the same input signals as their input terminals are tied together; hence, a single differential pair as input stage may be considered while internal current scaling may be implemented to define two output currents, one with a multiplication factor k with respect to the other. This scheme is shown in Fig. 3a and previously reported in [8]. Transistors M1-M5 form the differential input stage; M6-M7, and M8-M9,

form the output stages that provide currents i_o , and its scaled replica ki_o , respectively. This implementation eliminates the need for one of the differential input stages compared to circuit in Fig. 2b, thus reducing silicon area, and power consumption. Fig. 3b shows the circuit configuration that offers the equivalent operation of that in Fig. 2b. Unfortunately, this topology presents important disadvantages. Silicon area and current consumption in transistors M8-M9 increase proportionally to factor k . The drain resistance in these devices is inversely proportional to their biasing current due to the relation $r_o = 1/\lambda k I_B$; therefore, large values of k result in an important reduction of this value and, hence, the PLR of the multiplier, also reducing the frequency operational range.

Additionally, note that capacitor C_c forms an equivalent capacitance at node 'x' of value $C_x \approx C_c(1 + g_{m6}(r_{o6}||r_{o7}))$, defining the same dominant pole for the two outputs v_o , and v_{ok} ; however, v_{ok} has an open loop gain k times larger than v_o , thus shifting its unity-gain frequency to higher values, and causing possible stability problems. This pole has a low frequency effect on the transconductance at v_o of the form $g_{mo} = (g_{m6}g_{m2,3}R_x)/(1 + C_x R_x)$. At high frequencies the impedance from circuit in Fig. 3b, considering the ESR only, is defined as $Z = (g_{mo} + sC)/sCg_{mo}$, which combined with the previous relation results in complex poles, offering an undesired behavior compared to the typical ESR, as shown by simulation results in Section III.

B. Proposed Capacitor Multiplier

The proposed circuit is shown in Fig. 4a. It consists of a single differential pair formed by transistors M1-M14, which represents an advantage over circuit in Fig. 2a, as it does not have this stage, and over circuit in Fig. 2b which has two of them that increase mismatched effects and, hence, offset voltage. This stage gives an important property as explained later. Additionally, section marked as "Current multiplication" implements the scaling scheme included in circuit in Fig. 2a, thus, offering the same factor k with the advantage of including a differential input stage. This increases the multiplication factor exponentially to $k = 1 + k_1 k_2 k_3$ with reduced silicon area and power consumption compared to the implementations in Fig. 2b and 3a which only have one scaling stage for k . For the same scaling factor k , and assuming equal gain factors of $k_1 = k_2 = k_3 = k_x$ the proposed circuit has an addition in power consumption and silicon area proportional to the factor $3(k_x + 1)$, whereas for circuits in Fig. 2b and 3a this results in an exponential increment of k_x^3 . Also, the biasing current in the last stage of the proposed circuit, $k_3 I_B$, is reduced compared to case of that in Fig. 3a, $k I_B$. Therefore, the drain resistance in the output transistors, $r_o = 1/\lambda I_B$, is $k_1 k_2$ times larger for the proposed circuit, thus enhancing the PLR and the high-frequency corner of the operational range in the same magnitude. Note that these parameters are further enhanced by cascoding transistors. Additionally, dominant poles for outputs v_o and v_{ok} do not depend on the same node as in the case of Fig. 3a, which results in typical behavior from an ESR as

described previously and shown by simulations in Section III. Transistors marked as Msd are added in order to offer active source degeneration of transistors in the differential pair and increase the linearity compared to circuit in Fig. 3a, as well as to control the overall gain in order to ensure frequency stability as described in the next section.

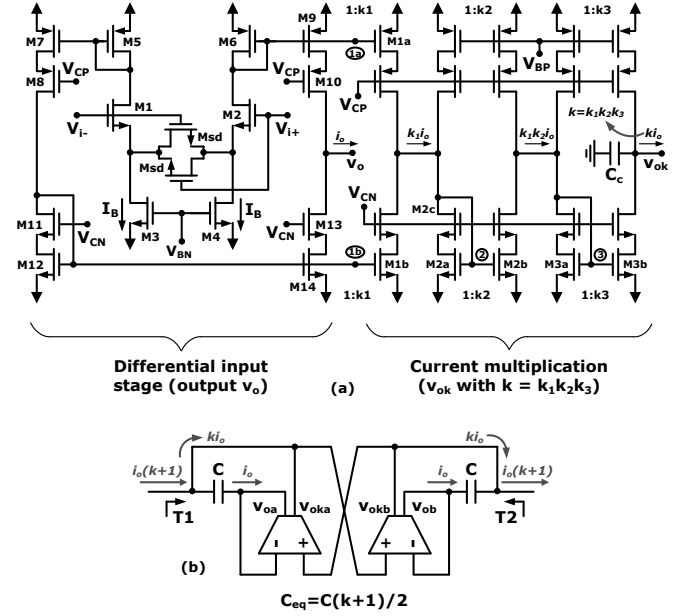


Fig. 4. a) Proposed differential capacitor multiplier, and b) circuit configuration to implement a floating capacitor multiplier.

The differential input stage offers an important advantage not present in circuits in [4]-[6]; it has the ability to define a voltage at both input terminals; if the non-inverting input of the implementation is connected to ground it operates as a grounded capacitor as in Fig. 3b. However, this input can be utilized to sense other voltage variations; thus, an additional terminal may be defined in order to operate the circuit as a floating capacitor. This structure is shown in Fig. 4b, where two replicas of the proposed circuit are required, one for each of the terminals. In this topology, the non-inverting input of each of the amplifiers is connected to the output node of the other multiplier; in this manner, each of the capacitors, C , is affected by the voltage at nodes $T1$ and $T2$. This induces the same current magnitude in both capacitors with opposite direction, which is then scaled by the corresponding amplifier at the output terminals. Note that cascoding is included in the current mirror of the output v_o due to the fact that in a floating implementation this node experiments considerable voltage swing. This reduces current variations in the differential pair that achieve output voltage adjustments.

C. Frequency Analysis

Considering the grounded configuration of the proposed circuit, as in Fig. 3b, a signal current induced in capacitor C is absorbed by terminal v_o and given by $i_c = i_{do} + i_{ro} + i_{Ci-}$, where i_{do} is the current defined by the transconductance gain at v_o , i_{ro} the current in the output resistance at this terminal, and i_{Ci-} the

total capacitance at inverting terminal. Also, considering an output current described as $i_{oTot} = i_c + i_{dok} + i_{rok}$, where i_{dok} is the current defined by the transconductance gain at v_{ok} , and i_{rok} the current in the output resistance at this terminal we obtain an equivalent impedance of:

$$Z_o = \frac{s(C + C_{i-}) + g_{mo} + 1/r_o}{\left(sC + 1/r_{ok}\right)\left(s(C + C_{i-}) + g_{mo} + 1/r_{ok}\right) + sC(g_{mok} - sC)} \quad (1)$$

Where g_{mo} and g_{mok} are the transconductance gain at v_o and v_{ok} respectively. Assuming small values of C_{i-} , and neglecting r_o due to its parallel combination with the low resistance $1/g_{mo}$, a LHP pole and zero are obtained at frequencies:

$$\omega_z \approx \frac{g_{mo}}{C}; \omega_p \approx \frac{g_{mo}}{C(1 + r_{ok}(g_{mo} + g_{mok}))} = \frac{1}{Cr_{ok}(k+1)}$$

Considering that the differential pair includes source degeneration the transconductance gain, and equivalent zero, result in:

$$g_{mDP} = \frac{4\beta_{1,2}\beta_{sd}\sqrt{I_B}}{(\beta_{1,2} + 4\beta_{sd})\sqrt{\beta_{1,2}}}; \omega_z \approx \frac{4\beta_{1,2}\beta_{sd}\sqrt{I_B}}{C(\beta_{1,2} + 4\beta_{sd})\sqrt{\beta_{1,2}}}$$

Where $\beta_{1,2}$ and β_{sd} represent the term $(1/2)\mu C_{ox}(W/L)$ of transistors in the differential pair, M1 and M2, and degenerating transistors Msd, respectively. In the current scaling block, the first stage provides a current $i_{o1} = k_{i_o} = g_{m1a,1b}(v_{gs1a} - v_{gs1b})$ by means of M1a and M1b, scaled by factor k_1 . In the next stage, the gate-source voltage of the diode connected transistor, v_{gs2a} , is induced according to:

$$v_{gs2a} = \frac{i_{i2}[r_{o2c} + r_{o2a}(1 + g_{m2c}r_{o2c})]}{1 + r_{o2a}(1 + g_{m2c}r_{o2c})(g_{m2a} + sC_2) + sC_2r_{o2c}} \approx \frac{1}{g_{m2a} + sC_2}$$

where C_a is the capacitance at node 'a'. An equivalent equation results at node 'b', thus, the transconductance gain of this stage, and equivalent pole of the overall system, are:

$$\frac{i_{ok}}{v_{gs1a} - v_{gs1b}} = \frac{g_{m1a,1b}g_{m2b}g_{m3b}}{(g_{m2a} + sC_2)(g_{m3a} + sC_3)}; \omega_p \approx \frac{1}{Cr_{ok}(k+1)}$$

Where, again, r_{ok} is the cascoded output resistance at v_{ok} . Regarding frequency stability, note from Fig. 3b that the two outputs form a feedback path with the inverting input, v_o directly, and v_{ok} by means of C . Although, v_{ok} presents an open-loop gain approximately k times larger than that at v_o , hence the former is studied for stability purposes. Large values of k increase the overall gain, introducing considerable phase shift above the unity gain frequency which may lead to instability, hence different methods may be implemented to prevent this. Transistors marked as Msd are included to reduce the transconductance gain of transistors in the differential pair, $g_{m1,2}$.

Additionally, a compensating capacitor C_c is also included in order to shift the dominant pole towards lower frequencies, although it increases the silicon area. Therefore, a combination of these two methods is used to achieve a tradeoff between silicon area and operational frequency. The open-loop dc gain at v_{ok} is defined as $A_{vk} = kg_{m1,2}r_{ok}$. Considering the dominant pole is located at the output node due to its large resistance, it is defined as $\omega_{p1} = 1/R_{ok}C_c$. The non-dominant internal pole

with the largest time constant is located at node '1a', due to the largest capacitance considering scaled PMOS transistors; hence a pole is defined at $\omega_{p2} = g_{m6}/C_{1a}$, where g_{m6} is the transconductance gain of M6 and C_{1a} the parasitic gate capacitance at that node. This leads to a gain-bandwidth product given by $GB = A_{vk}\omega_{p1} = kg_{mDP}/C_c$. Typically the condition $2GB < \omega_{p2}$ is employed to ensure frequency stability, therefore the condition $g_{m6}/g_{mDP} = 2k(C_{1a}/C_c)$ must be met.

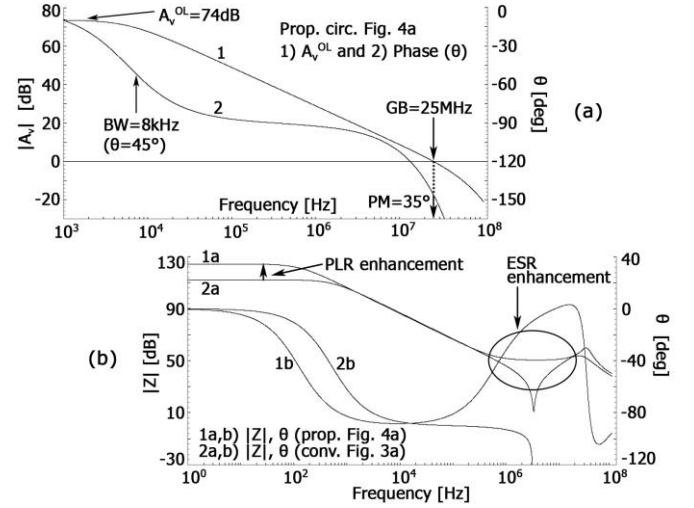


Fig. 5. a) Open-loop frequency response of circuit in Fig. 4a, and b) Comparison of impedance between circuit in Fig. 3a and proposed circuit in Fig. 4a, shown in magnitude and phase.

III. SIMULATION AND EXPERIMENTAL RESULTS

The proposed operation was simulated using spectre in CMOS technology. Fig. 5a shows the frequency response of the open-loop operation of the proposed circuit at output v_{ok} . Fig. 5b shows a comparison of the simulated impedance between circuits in Fig. 3a and 4a. Observe that the proposed circuit has an increased PLR and a typical behavior of the ESR compared to the conventional circuit.

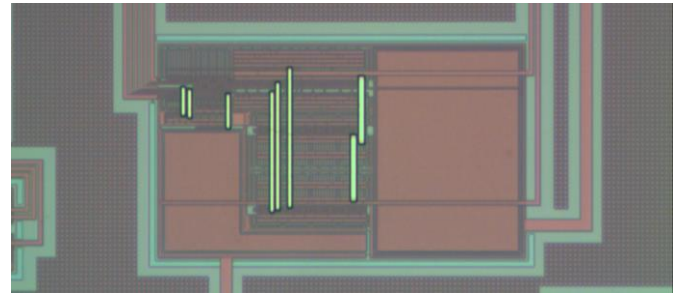


Fig. 6. Microphotograph of the fabricated circuit (385 μm x 225 μm).

Also, a prototype circuit was fabricated in On-Semiconductor 0.5 μm double-poly three-metal CMOS technology through MOSIS. Fig. 6 shows the microphotograph of the circuit. The nominal threshold voltages are $V_{THN} = 0.7$ V and $V_{THP} = -0.9$ V and the transistor sizes are 46.8/1.2 and 126.6/1.2 $\mu\text{m}/\mu\text{m}$ for NMOS and PMOS respectively (Msd are 15/4.8 $\mu\text{m}/\mu\text{m}$). The circuit was tested with a supply voltage of $V_{DD} - V_{SS} = 2.2$ V and a biasing current $I_B = 40$ μA , resulting in

a saturation voltage $V_{DSsat} \approx 0.15$ V. The values of the capacitors are $C = 25$ pF and $C_c = 13$ pF. A scaling factor of 3 was used in all stages, resulting in a gain factor of $k=(1+k_1k_2k_3)=28$. With these values the target parameters are $C_{eq} = 700$ pF, PLR = 1.58 M Ω , ESR = 347 Ω , a pole at $f_p = 131$ Hz, and a zero at $f_z = 626$ kHz. Fig. 7 shows the experimental impedance of the proposed circuit. The measured parameters are $C_{eq} = 753$ pF, considering ~ 50 pF of parasitic capacitance due to the testing board, a pole at $f_p = 153$ Hz, a zero at $f_z = 478.63$ kHz, and the PLR and ESR at 1.28 M Ω and 392 Ω respectively, which are in good agreement with theoretical values. Fig. 8a shows the measured output voltage of the grounded and floating configurations using the proposed circuit in Fig. 4a, implemented in a low-pass and a high-pass RC filter respectively. They both have a base capacitance of 80pF, resulting in an equivalent capacitance $C_{eq} = 2.24$ nF (again, including a parasitic capacitance of ~ 50 pF) combined with a resistor $R = 22$ k Ω . The measured cutoff frequencies are 3.3 and 3.1 kHz for the grounded and floating implementations respectively.

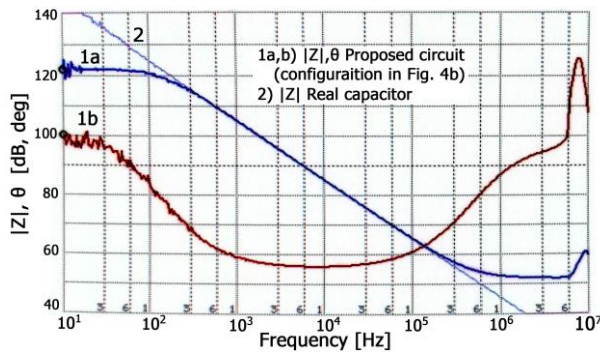


Fig. 7. Measured impedance of circuit in Fig. 4b.

Additionally, the grounded and floating realizations were implemented in a second-order unity-gain Tschebyscheff low-pass filter using the Sallen-Key topology; the experimental operation is shown in Fig. 8b. The transfer function is given by $A(s) = 1/(1+sRC)^2$, for the values $C = 80$ pF for a $C_{eq} = 2.225$ nF, and $R = 7.6$ k Ω , which result in a cutoff frequency of 9.332 kHz. Table I shows the measured results of the proposed circuit compared to structures previously reported. A larger multiplication factor with reduced silicon area and power consumption with respect to other topologies is shown.

IV. CONCLUSION

A capacitor multiplier based on a differential amplifier with exponential current scaling was presented. The reduction to a single differential input stage resulting in increased accuracy, and reduced silicon area and power consumption was discussed. The differential input stage also offers the capability to implement grounded and floating realizations. Experimental results showing the operation of the grounded and floating circuits implemented in low- and high-pass RC filters, and a second order low-pass filter, were presented.

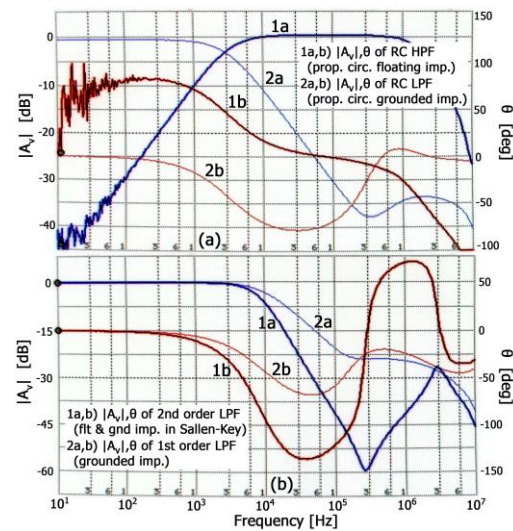


Fig. 8. a) Frequency response of an RC low- and high-pass filters using the grounded and floating configurations of circuit in Fig. 4a respectively; b) frequency response of a 2nd order Sallen-Key LPF filter using these configurations.

TABLE I
COMPARISON OF MEASURED PARAMETERS BETWEEN CONVENTIONAL CIRCUITS AND PROPOSED CIRCUIT IN FIG. 4A.

Parameter	[4] (2006)	[6] (2009)	[7] (2009)	[8] (2012)	Prop. Fig. 4a
Technology [μ m]	0.35	0.18	OTAs CA3280	0.5	0.5
Multiplication factor	16	65	22	10.1	28
Base capacitance C [pF]	170	7.95	-	18	25
C_{eq} [pF]	2720	516	-	182	700
ESR [Ω]	316	-	-	-	392
PLR [Ω]	1 M	17 M	1.5 M	-	1.28 M
Power consumption [mW]	0.672	-	-	-	1.32
Minimum supply voltage [V]	-	-	-	-	1.3
Silicon area [mm ²]	-	-	-	0.070 ~	0.07

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